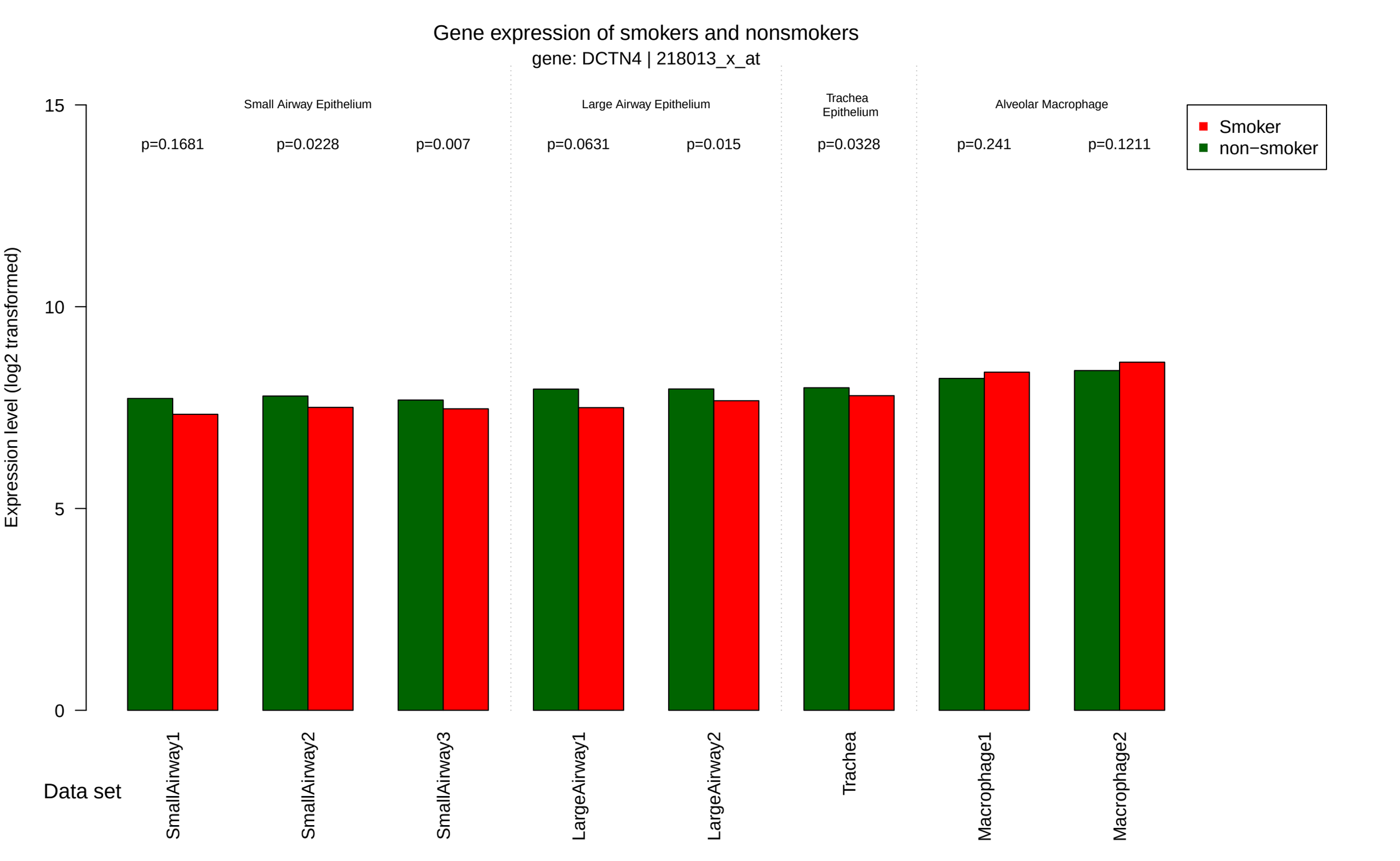
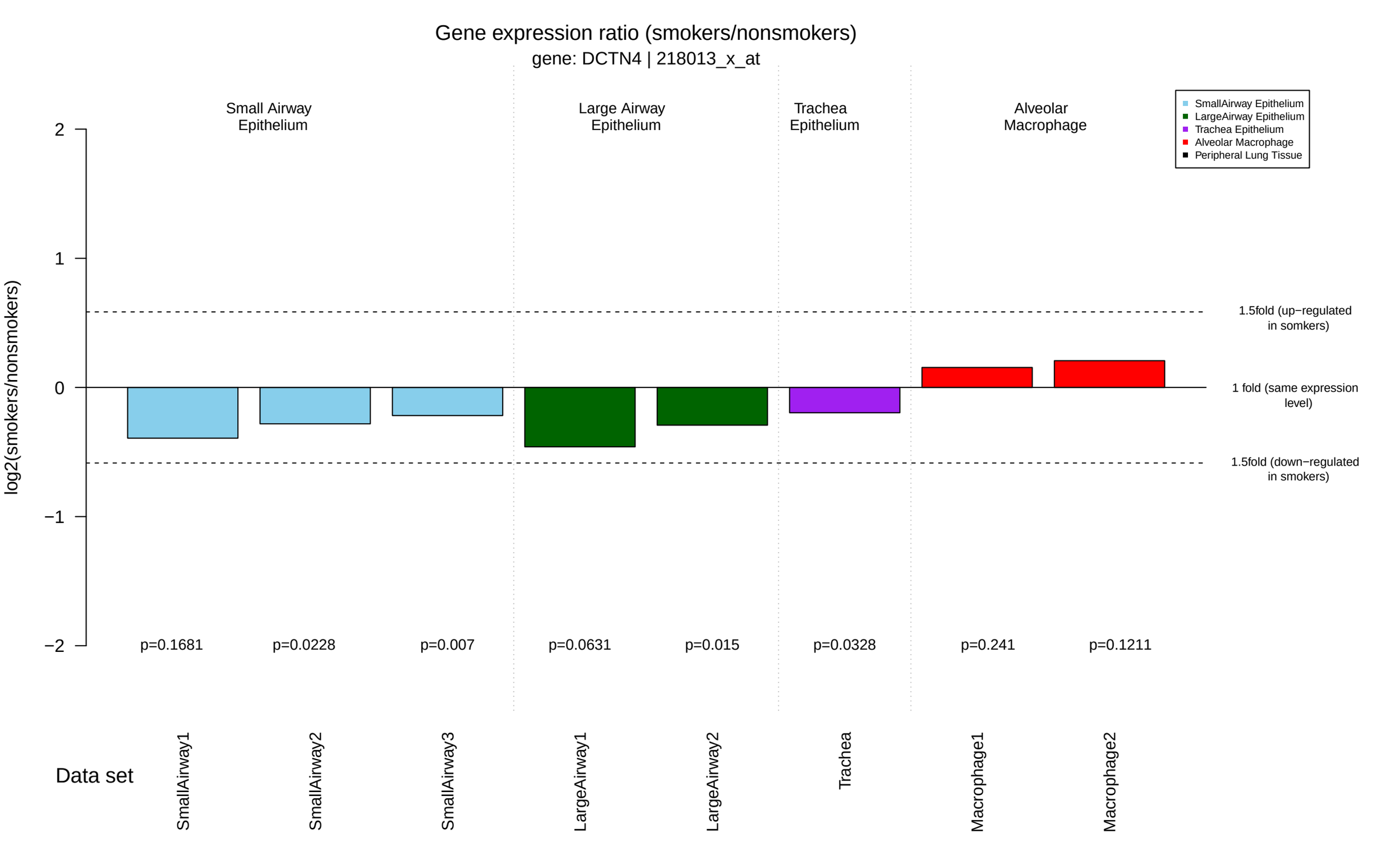
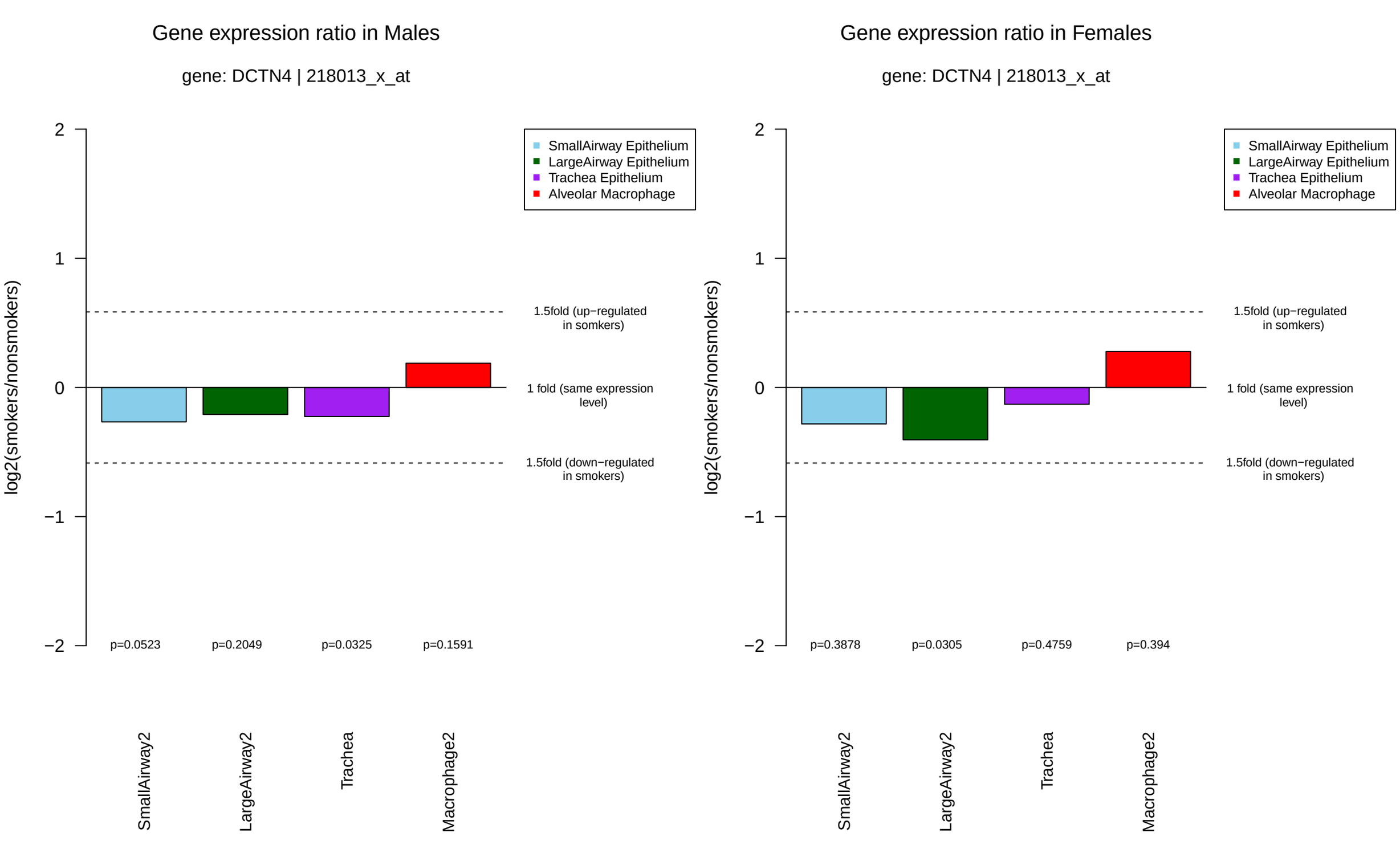
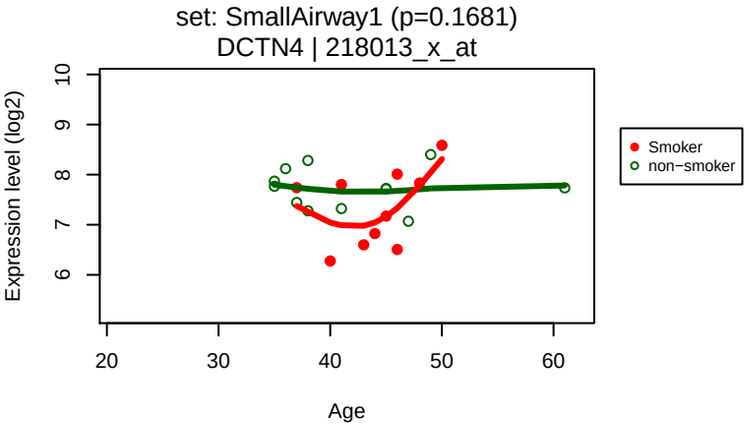
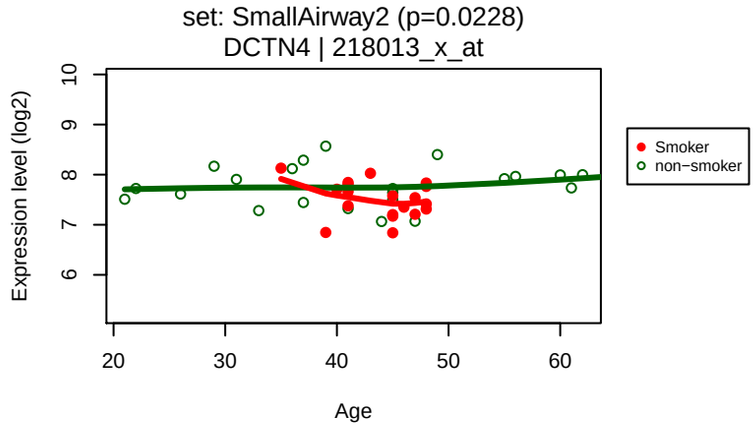
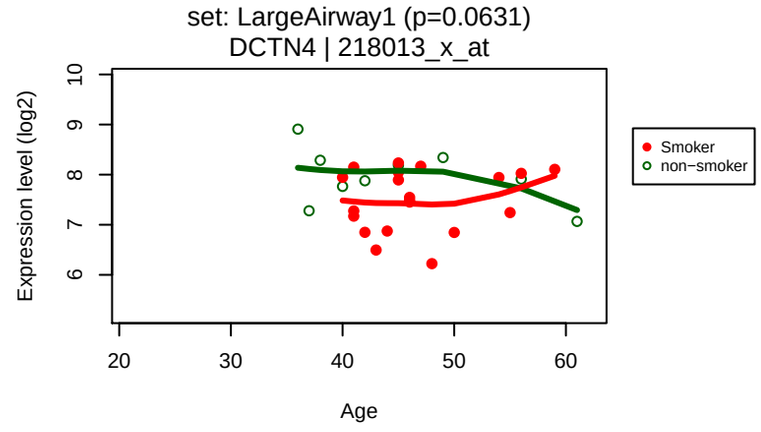

Gene expression of smokers and nonsmokers
gene: DCTN4 | 218013_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1681
p=0.0228
p=0.007
p=0.0631
p=0.015
p=0.0328
p=0.241
p=0.1211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

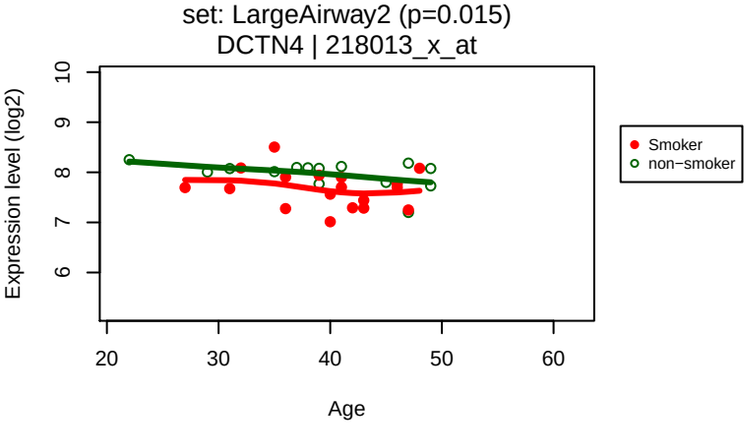

Gene expression ratio (smokers/nonsmokers)
gene: DCTN4 | 218013_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1681
p=0.0228
p=0.007
p=0.0631
p=0.015
p=0.0328
p=0.241
p=0.1211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

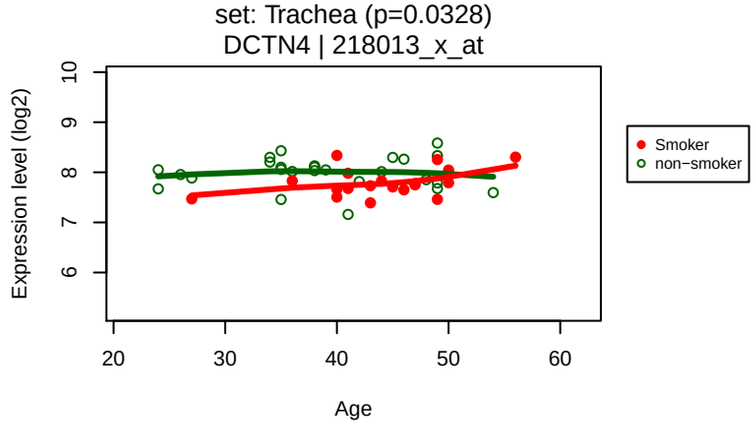

Gene expression ratio in Males
gene: DCTN4 | 218013_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0523
p=0.2049
p=0.0325
p=0.1591
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DCTN4 | 218013_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3878
p=0.0305
p=0.4759
p=0.394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

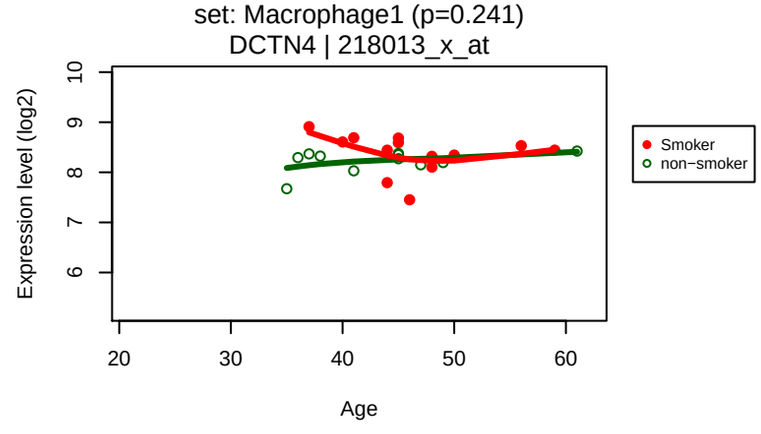

set: SmallAirway1 (p=0.1681)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker

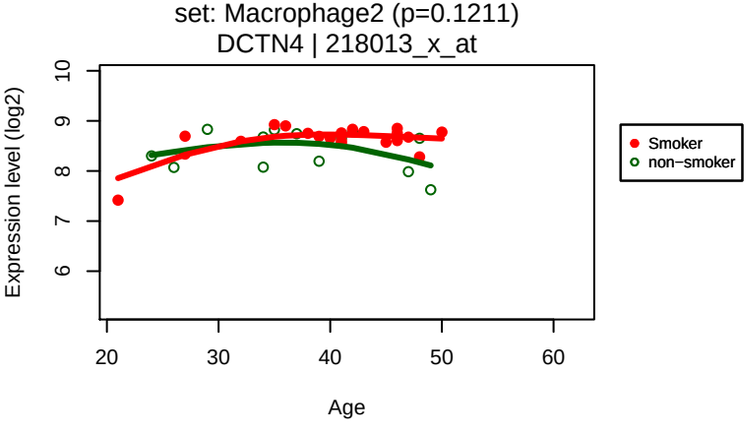

set: SmallAirway2 (p=0.0228)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker

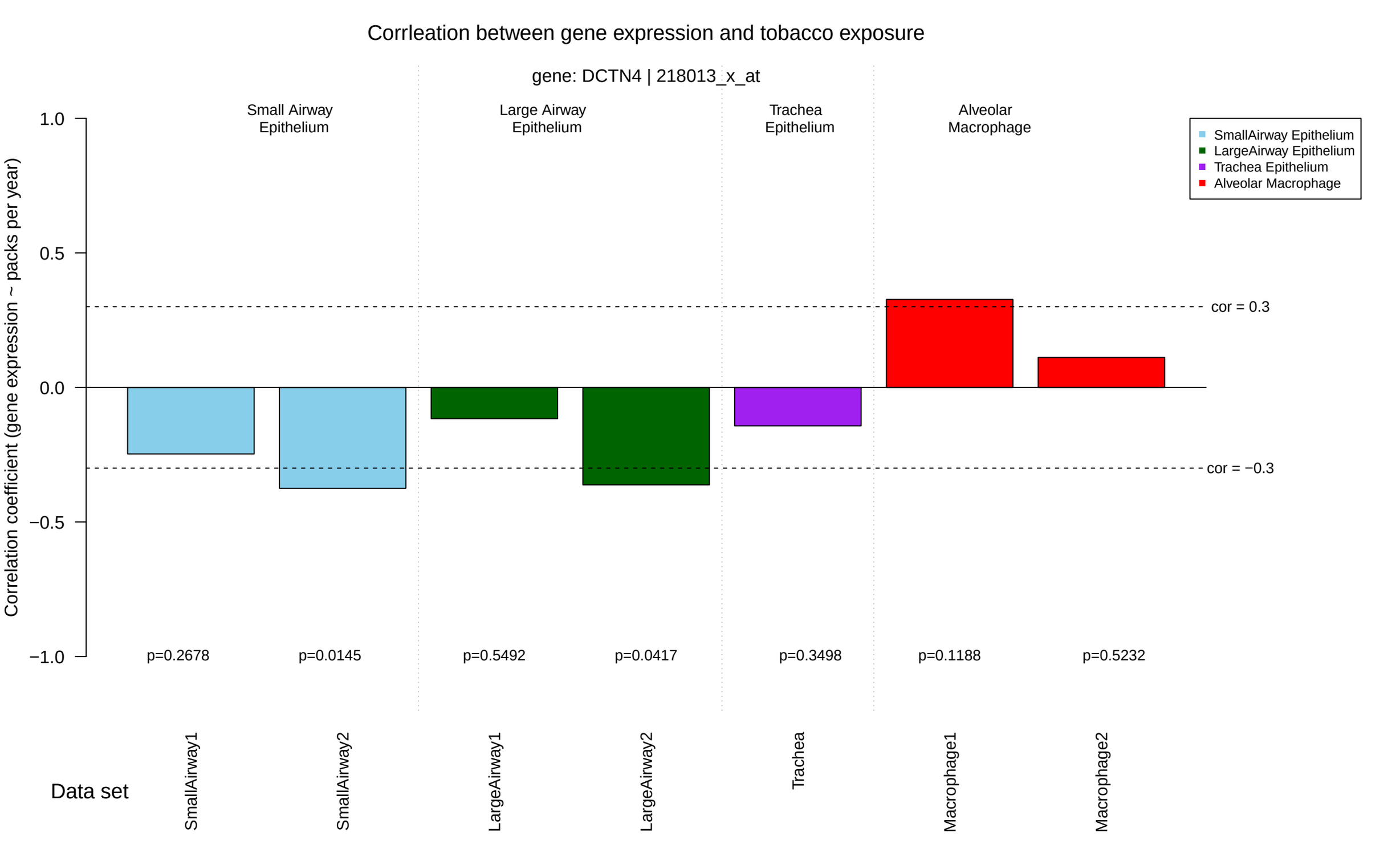

set: LargeAirway1 (p=0.0631)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.015)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0328)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.241)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1211)
DCTN4 | 218013_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DCTN4 | 218013_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2678
p=0.0145
p=0.5492
p=0.0417
p=0.3498
p=0.1188
p=0.5232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2